Geophys. J. Int. (1995) **120,** 433–444

A new Late Triassic–Liasic Palaeomagnetic pole from superimposed and juxtaposed magnetizations in the Saharan craton

B. Kies,[1,2] B. Henry,[1] N. Merabet,[3] M. M. Derder[3] and L. Daly[1]

[1] *GDR 'Magnétisme des Roches' du CNRS, IPGP and Paris* VI *University,* 4 *avenue de Neptune,* 94107 *Saint-Maur cedex, France*
[2] *Oran University,* BP 1524 *Oran, Algeria*
[3] *Centre de Recherche en Astronomie, Astrophysique et Géophysique (CRAAG),* BP 63, 16340 *Bouzareah, Alger, Algeria*

Accepted 1994 August 19. Received 1994 March 15; in original form 1993 August 5

SUMMARY

Palaeomagnetic measurements, in the Late Triassic–Liasic Zarzaïtine formations of the La Reculée area, showed the existence of superimposed and juxtaposed magnetizations. The separation of the juxtaposed magnetizations yielded Late Triassic–Liasic magnetization directions of normal and reversed polarities. The new corresponding palaeomagnetic pole agrees well with a Pangea A configuration. It confirms, from Autunian to Lias, a latitudinal displacement of stable Africa towards the north, with important anti-clockwise rotation contrary to Laurussia which did not undergo significant rotation. This transition of Pangea from A2 to A ended before the Late Triassic. It corresponds to dextral movements in the Hercyno-Appalachian collision area, and perhaps to part of the northwards drift of microcontinents of Gondwanian origin located north of the Arabian promontory.

Key words: palaeomagnetism, Pangea, Saharan craton, Triassic–Lias.

INTRODUCTION

Even though the Palaeozoic segment of the Apparent Polar Wander Path (APWP) of Africa has been improved by recent data (Henry, Derder & Merabet 1992a), the Lower Mesozoic segment appears still insufficiently specified. This is unfortunately due to widespread stratigraphic gaps in the Upper Permian–Middle Triassic series of stable Africa. On the other hand, the rare levels of this age are mostly remagnetized or unfavourable for palaeomagnetic analysis (Ghorabi & Henry 1991). Except in southern Tunisia (Busson 1972; Bouaziz 1986), non-saliferous facies exist mainly in the Illizi basin, in eastern Algerian Sahara. It was therefore of major interest to study the Zarzaïtine formations in this basin, which are the bottom of the Mesozoic series. A new Upper Triassic–Liasic palaeomagnetic pole should be a useful contribution to specify the Gondwanian motion up to the breaking up of Pangea. In fact, insufficient palaeomagnetic constraints on the age of the transition of Pangea from A2 to A are presently available.

GEOLOGICAL SETTING

The lower Zarzaïtine Formation (Busson 1972; Busson & Cornée 1989a, b) comprises predominantly coarse white to red sandstones, with some interbedded white or red clay levels. Vertebrate fauna, and in particular Stegocephalia (similar to *Stenosaurus*), Sauria (likely Archosauromorpha or Lepidosauromorpha—Jalil 1990) and fish (*Hybodus, Ceratodus*), give an Upper Triassic–Rhaetian age (Lehman 1971) for the basal levels; palynological data also indicate the same levels for Carnian age, with a *Camerosporites secatus* flora association (Achab 1970). The Middle Zarzaïtine Formation, dolomitic sandstones and sandstones, is attributed to the Lias–Dogger (Busson & Cornée 1989b). This dating is based, on the one hand, on the existence of the dated Upper Triassic rocks at the bottom of the Zarzaïtine Formation, of Lower Liasic flora (*Classopolis*) observed in a bore hole at the top of the Lower Zarzaïtine (Achab 1970) and of the dated Upper Jurassic level in the overlying Taouratine Formation; on the other hand, this dating is supported by correlations with series from northern and central Sahara (Busson 1972; Busson & Cornée 1989a,b).

In the La Reculée area, the Lower Zarzaïtine constitutes the La Reculée cliff, but the Middle Zarzaïtine outcrops are located only in the basal levels of the dolomitic and detritic facies attributed to the Lias (Busson & Cornée 1989b).

From a structural point of view, the Zarzaïtine beds mainly appear as tabular formation, horizontal or subhorizontal. However, in the La Reculée area, some tilting is related to north–south faults, and moderate dip variations have been observed. In the south of the Tiguentourine Life Base of the SONATRACH, a small

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structural basin (apparent dip locally reaching 21°) of Stegocephalia coarse sandstones has been observed within the Upper Tiguentourine outcrops.

SAMPLING AND ANALYSIS PROCEDURE

In facies favourable to palaeomagnetic investigations, 144 orientated samples were collected from 14 sites (Fig. 1) distributed all over the La Reculée area (27.9°N, 9.3°E).

Six of these sites belong to the Upper Triassic beds. Samples from one (site 9) of these sites comprise coarse red sandstones of the level with Stegocephalia; sampling was done, despite the coarse nature of the rocks, considering the local apparent dip of 21° (useful to attempt a fold test). Three sites (1, 3 and 7) yielding only hand-samples belong to an upper red clay bed (Ta2 clay in Busson 1972; Busson & Cornée 1989a,b). Finally, samples from the last two sites (8 and 14) are composed of fine-grained sandstones, which are situated close to the top of the Late Triassic beds.

The remaining eight sites have a Liasic age, and samples are composed, except fine sandstones for site 2, of dolomitic sandstones collected at the bottom of the carbonated facies.

Standard cylinder specimens were cut in the laboratory from each sample, except for the 27 swelling clay handsamples; these last samples were too friable to be drilled in the laboratory, even using forced air instead of water, and moreover could not be consolidated by aqueous sodium silicate solution because of their swelling facies. These 27 hand samples therefore were rejected.

Prior to any demagnetization analysis, the specimens were stored in a zero-magnetic field for at least 1 month to obtain a decay of Viscous Remanent Magnetization (VRM)

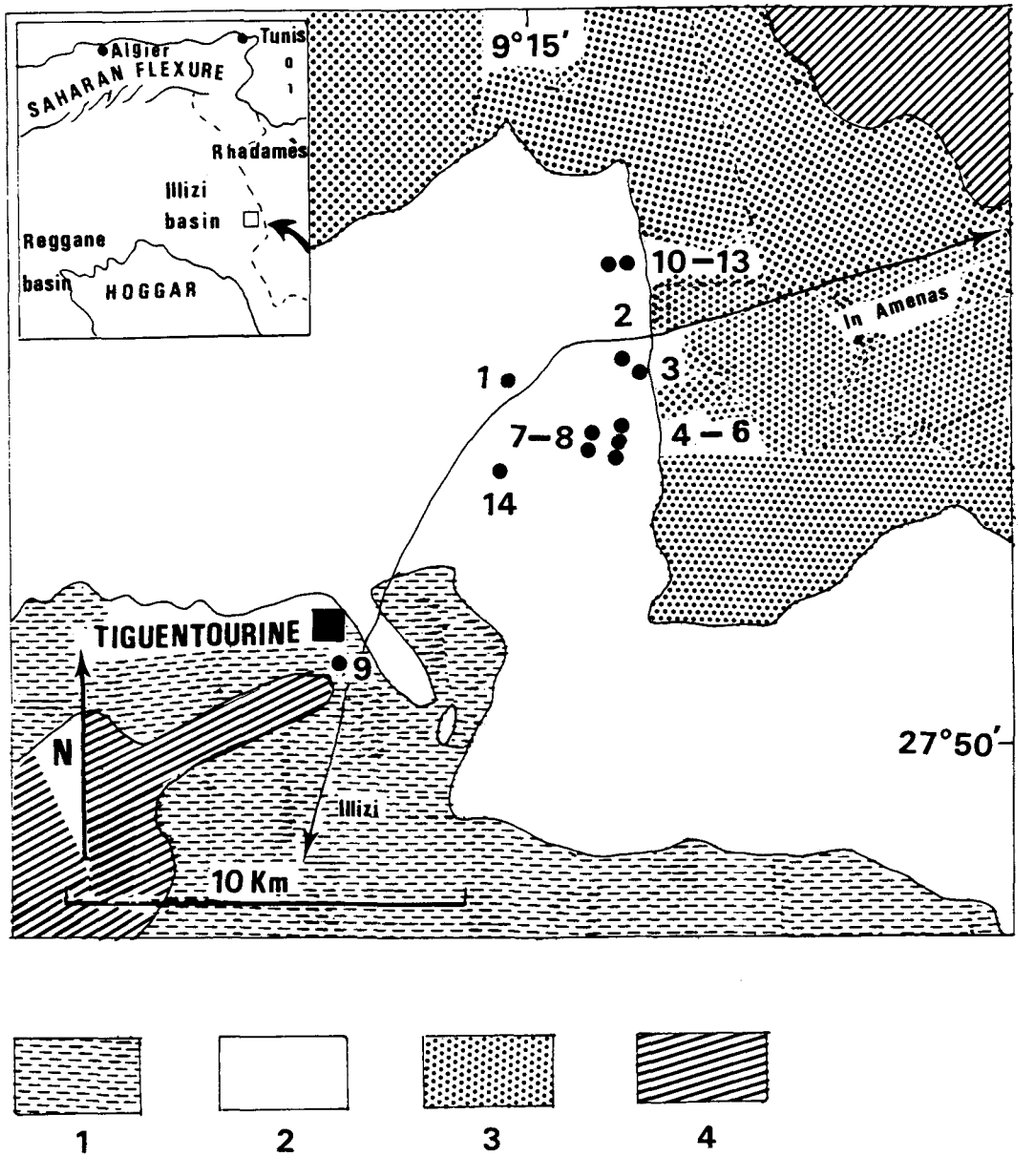


Figure 1. Geological map (after Bonnet, Dubois & Busson 1965) of the studied area and sampling sites (●, with site no.): 1, Upper Tiguentourine; 2, Lower Zarzaïtine; 3, Middle Zarzaïtine; 4, Erg.

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component acquired *in situ* or after sampling. The Natural Remanent Magnetization (NRM) was measured using a JR-4 spinner magnetometer (Geofyzika, Brno). Thermal demagnetization was performed using a cylindrical furnace, with a fast heating rate, and cooling by forced air at the heating site. Selected thermal increments were 100° to 200 °C treatment, and 50–20 °C for higher temperatures. Reduced temperature increments were used at the highest temperatures in order to obtain a precise determination of the magnetization evolution. Alternating field treatments up to 160 mT were attempted using an apparatus which automatically cancels the parasitic anhysteretic effect (Le Goff 1985), but blocking field values in these specimens were too high to permit any significant demagnetization by this means.

Orthogonal vector plots (Wilson & Everitt 1963; Zijderveld 1967), Fisher's statistic (1953) and bivariate distributions (Le Goff 1990; Le Goff, Henry & Daly 1992) providing a confidence ellipse around a vectorial weighted mean, were used for data analysis.

ROCK-MAGNETIC AND PALAEOMAGNETIC RESULTS

The NRM intensity values σ are weak (average value $3.5 \times 10^{-7}\,Am^2\,kg^{-1}$) and relatively homogeneous in all samples. The magnetic susceptibility K of some pilot specimens were measured, at room temperature after each heating step, using Kappabridge KLY-2 (Geofyzika, Brno). A weak decrease of K has been observed between the 200° and 400 °C treatments, mainly in dolomitic facies (probably due to oxidation of the low amount of ankerite found in these facies), and a strong increase after heating at temperatures higher than 450 °C. In order to specify the magnetic mineralogy, and its evolution during thermal treatment, hysteresis loops have been performed before and after heating at 700 °C (Fig. 2) using a translation inductometer within an electromagnet providing a field up to 1.6 T. The hysteresis curves are very different. Before heating (Fig. 2a), there is no saturation, even in high field, and the remanent coercive force is high ($H_{cr} = 0.3$ T). This indicates the presence of haematite. After heating (Fig. 2b), the susceptibility in high field is five times more important, and saturation is reached at a field lower than 0.2 T; the coercive force is low ($H_{cr} = 0.06$ T); this indicates the formation of mineral(s) of low coercivity and disappearance, at least for a part, of haematite during thermal treatment. An attempt at Curie curve determination from an unheated sample gave insignificant results for the heating curve because of the very low mean susceptibility values. However, the cooling part of the curve is better, because of the strong susceptibility increase. This indicates the presence of minerals with a Curie temperature higher than 400 °C and lower than 600 °C. Magnetite is then probably the result of the mineralogical change due to heating at temperatures higher than 600 °C.

The NRM directions are relatively close to the present-day field (Fig. 3). Although NRM is carried by haematite with high remanent coercive force, blocking temperatures are mostly lower than 600 °C (often lower than 500 °C), and therefore can be wholly analysed without the occurrence of the disturbances related to the mineralogical change.

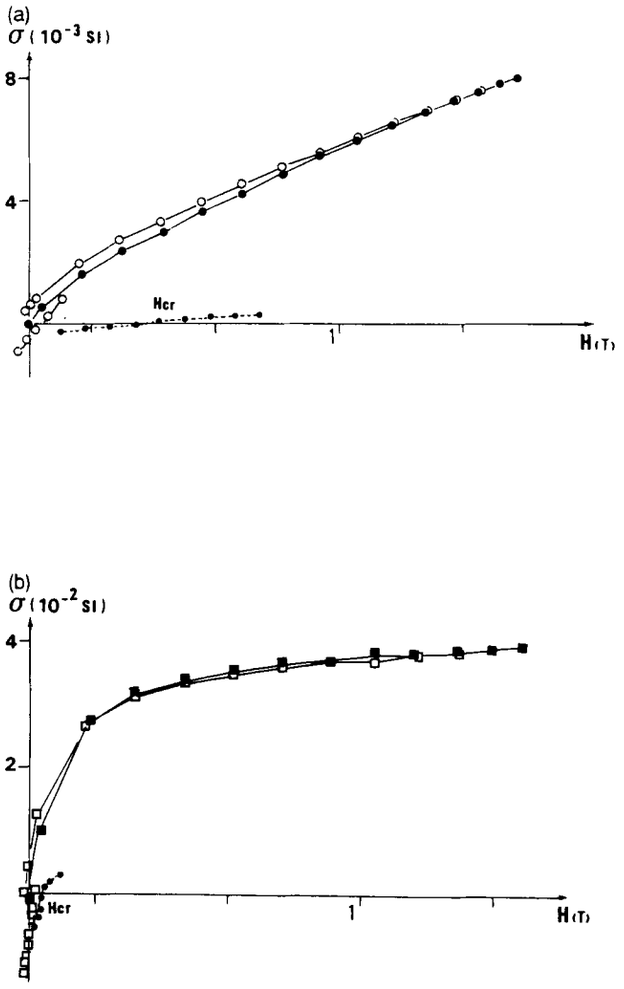


Figure 2. Hysteresis loops of the sample Za8 before (a) and (b) after heating at 680°C; H_{cr} is the remanent coercive force.

Thermal treatments showed, after elimination of the VRM, different types of magnetic behaviour.

No stable magnetization directions

A first type, characterized by superimposed components (Fig. 4a), was observed in 11 of the 21 samples from site 9 (coarse sandstones). Indeed, on an equal area projection plot, the magnetization direction varies along a great circle path during the different thermal treatment steps. The 11 great circles have a well-defined (Henry, Merabet & Bourouis 1992b; Derder *et al.* 1994) intersection: before dip correction, $D = 4°$, $I = 37°$, $\alpha_{95} = 1.2°$ according to the McFadden & McElhinny (1988) method. This direction is close to that of the present magnetic earth field and also to the mean of poorly specified Characteristic Remanent Magnetization (ChRM) of 10 other samples belonging to this site (before dip correction, $D = 1°$, $I = 45°$). Moreover, for the 11 samples allowing great circle determination, the evolution of the magnetization direction towards a southerly

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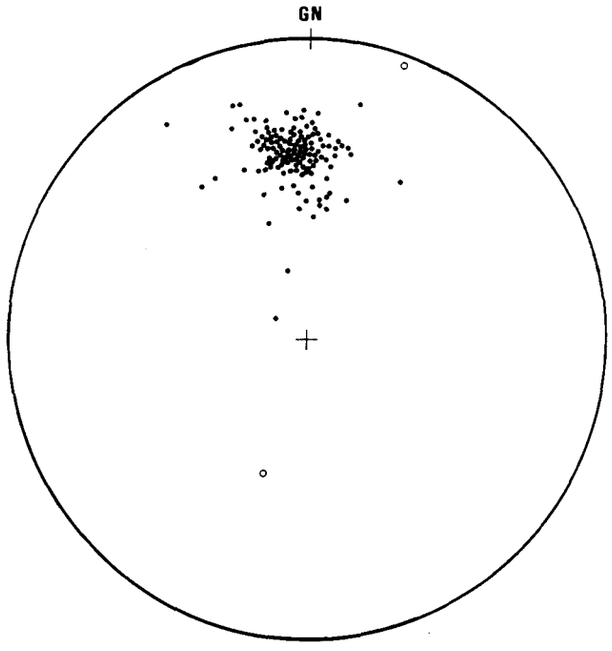


Figure 3. Equal area plot (filled symbols, lower hemisphere; open symbols, upper hemisphere) of NRM directions before dip correction.

orientation at the highest temperature indicates that the other component is of reversed polarity.

ChRM directions

A well-defined ChRM (Fig. 5) with a high blocking temperature spectrum and normal polarity has been identified in numerous samples (81 samples from 11 sites): $D = 349.2°$, $I = 30.5°$, $k = 50$, $\alpha_{95} = 2.2°$ and $D = 348.4°$, $I = 32.0°$, $k = 36$, $\alpha_{95} = 2.6°$, respectively, before and after dip correction. However, Fig. 5 shows a scattering in partial girdle of these ChRM directions; bivariate analysis (Le Goff 1990; Le Goff *et al.* 1992) of the distribution gives ellipticity ratios k_y/k_x of 5.1 and 2.4, respectively, before and after dip correction; these values are characteristic of non-circular distributions for 81 directions (Le Goff *et al.* 1992; Henry & Le Goff 1994). Such a distribution indicates that this ChRM probably corresponds to at least two different components of magnetization.

In addition to the mentioned ChRM normal component, 44 (eight sites) of these 81 samples, are characterized by the existence of a low blocking temperature component (Fig. 6), of normal (**group a**) or reversed (**group b**) polarity according to the samples (juxtaposed magnetizations—Daly 1981). The other 37 samples only show a single component of magnetization (**group c**).

Group a. 21 of these 44 samples exhibited two components with normal polarity (Fig. 4b). However, their directions are clearly different. The second component (squares in Fig. 6), slightly different to the VRM component related to the present field, corresponds to a relatively low blocking temperature mostly of the order of 250°–300 °C. The mean directions for this second component ($N = 4$ sites) are $D = 2.3°$, $I = 37.9°$, $k = 547$, $\alpha_{95} = 3.0°$, $k_y/k_x = 20.3$, and $D = 359.2°$, $I = 42.0°$, $k = 92$, $\alpha_{95} = 7.3°$, $k_y/k_x = 8.3$, respectively, before and after dip correction (Fig. 6). The fold test is clearly negative according to Fisher's parameter. High ellipticity values are statistically characteristic of non-circular distribution for four data, however the ratio k_y/k_x before dip correction has possibly no meaning because of the high k_y value (Henry & Le Goff 1995).

Group b. The remaining 23 samples were all collected in dolomitic sandstones levels. After elimination (150°–200° C heating) of a strong VRM component of normal polarity and certainly connected to the present magnetic field, a second component, of reversed polarity (circles in Fig. 6), with a relatively low blocking temperature spectrum (300°–350 °C) can be identified (Fig. 4c). It is quite anti-parallel to the normal polarity ChRM of the same samples (angular deviation of 0.6° between mean directions of normal and reversed components). The mean directions of these 23 reversed directions ($N = 4$ sites) are $D = 164.5°$, $I = -25.7°$, $k = 296$, $\alpha_{95} = 4.1°$, $k_y/k_x = 4.0$, and $D = 166.3°$, $I = -25.7°$, $k = 212$, $\alpha_{95} = 4.8°$, $k_y/k_x = 1.8$, respectively, before and after dip correction (Fig. 6). The fold test yields no results using Fisher's parameters that can be used; both k_y/k_x values are too low to indicate a significant ellipticity for four directions (Le Goff *et al.* 1992; Henry & Le Goff 1995).

Comparison between the normal polarity ChRM (high blocking temperature spectra) directions of **groups a** and **b** (circles in Fig. 5), and reversed ones (low blocking temperature spectra) of **group b** (circles in Fig. 6) shows that they are very close to each other (angular deviation of 0.5° between mean directions of normal and reversed components). Then a mean direction can be determined by using all these directions ($N = 8$ sites, $D = 344.8°$, $I = 25.7°$, $k = 423$, $\alpha_{95} = 2.4°$, $k_y/k_x = 4.6$, and $D = 346.0°$, $I = 26.1°$, $k = 355$, $\alpha_{95} = 2.6°$, $k_y/k_x = 1.4$, respectively, before and after dip correction). A very clear positive inversion test is obtained by comparison of the normal and reversed directions; it indicates the absence of another component. The fold test is insignificant using Fisher's parameters, but positive considering the decrease (from a value characteristic of a non-circular distribution to values compatible with a circular distribution hypothesis, for eight directions) of ellipticity ratio during dip correction (Henry & Le Goff 1994).

A surprising fact arises from comparison of *low-temperature* components of **groups a** (normal) and **b** (reversed). Indeed, these components have neighbouring blocking temperature spectra, but they correspond to different directions (angular deviation of 19.4° between their mean directions). Therefore, this means different ages of magnetization acquisition. Another great difference between these two groups is related to the dip correction effect; the fold test is positive for reversed directions, while it is clearly negative for the normal directions (the high ellipticity of the normal group, before as after dip correction, might indicate that the 'single' component of this group corresponds in fact to superimposed magnetizations).

Group c. The last type of magnetic behaviour is illustrated by Fig. 4(d), where only a single remanent magnetization component constitutes (with the VRM) the NRM of 37 samples: $D = 358.9°$, $I = 37.3°$, $k = 67$, $\alpha_{95} = 2.6°$, $k_y/k_x = 2.6$, and $D = 356.4°$, $I = 39.5°$, $k = 32$,

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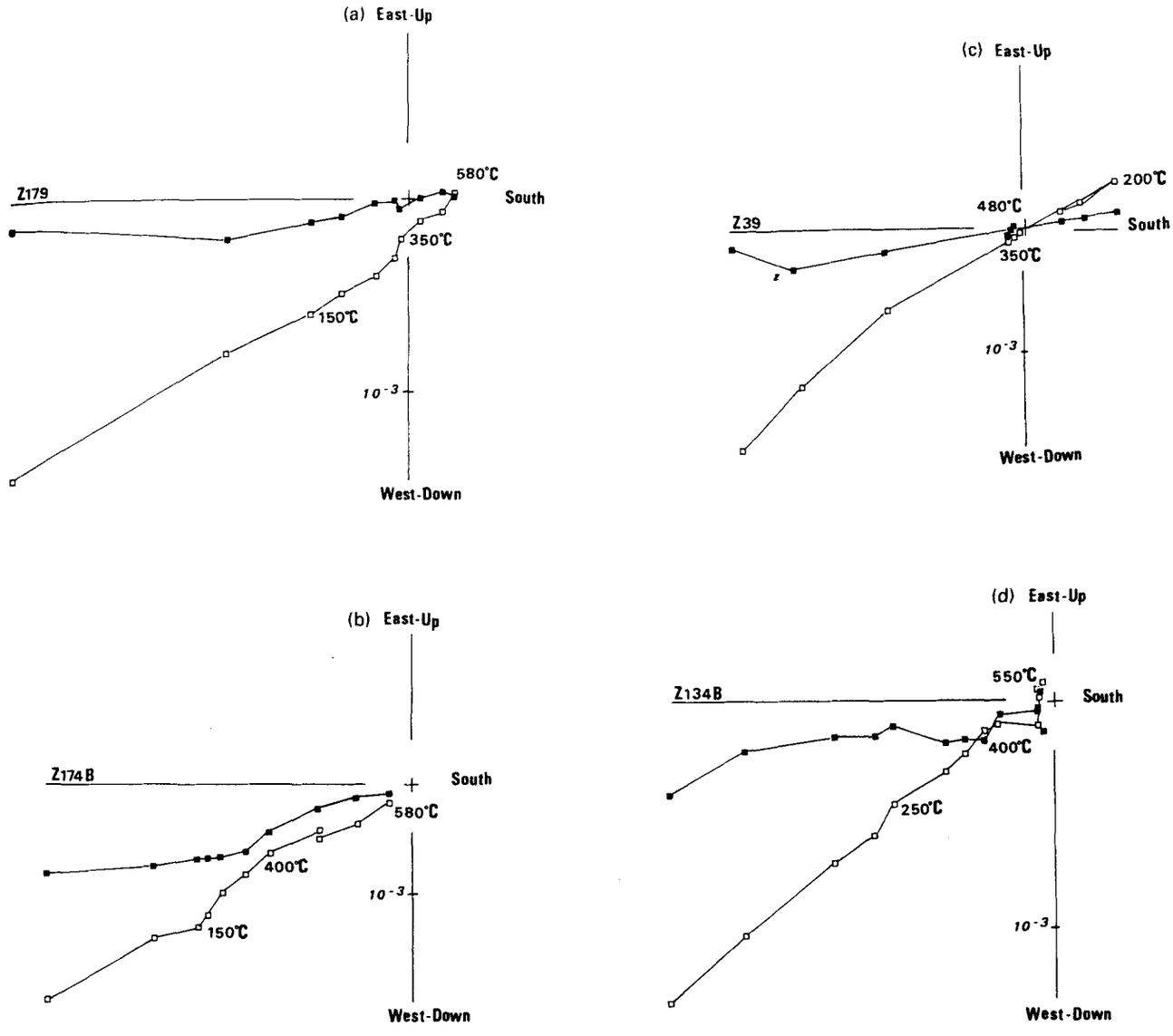


Figure 4. Orthogonal vector plots with heating value in degrees Celsius (filled symbols: horizontal plane; open symbols: vertical plane; intensity in A m^{-1}).

$\alpha_{95} = 3.8°$, $k_y/k_x = 3.3$, respectively, before and after dip correction. The fold test is negative according to Fisher's and ellipticity parameters, but k_y/k_x values are characteristic of a non-circular distribution. This **group c** of directions then includes superimposed magnetizations (at least the main component being a post-tilting overprint). Several of these samples showed magnetization directions close to the second normal one of the **group a,** corresponding perhaps to a true single component. But, most of the directions are in intermediate positions between the two component directions of **group a.** The plane associated with lengthening the confidence ellipse of the ChRM distribution of **group c** also includes these two component directions of **group a.** This is complementary evidence of the superimposition of these components (Henry & Le Goff 1994; their separation is practically impossible even using thermal or alternating field treatments (Daly 1981). Nevertheless, these last 37 samples have been rejected.

DISCUSSION

Most of the sites are situated along two cross-sections of the Late Triassic–Liasic series and are arranged successively from the bottom to the top as follows: 7, 8, 5, 6, 4 in the first section, and 10, 11–13 in the second one. Sites 7, 11 and 12 did not yield reliable results. Despite their different stratigraphic levels, all the samples from sites 8, 5 and 6 of the first section present both normal and reversed polarities, while those of site 4 in the top level contain only normal polarity. In the second section, only normal polarity has been identified in site 10, and both polarities in the site 13.

Sites 1 and 3 from other areas did not yield reliable results

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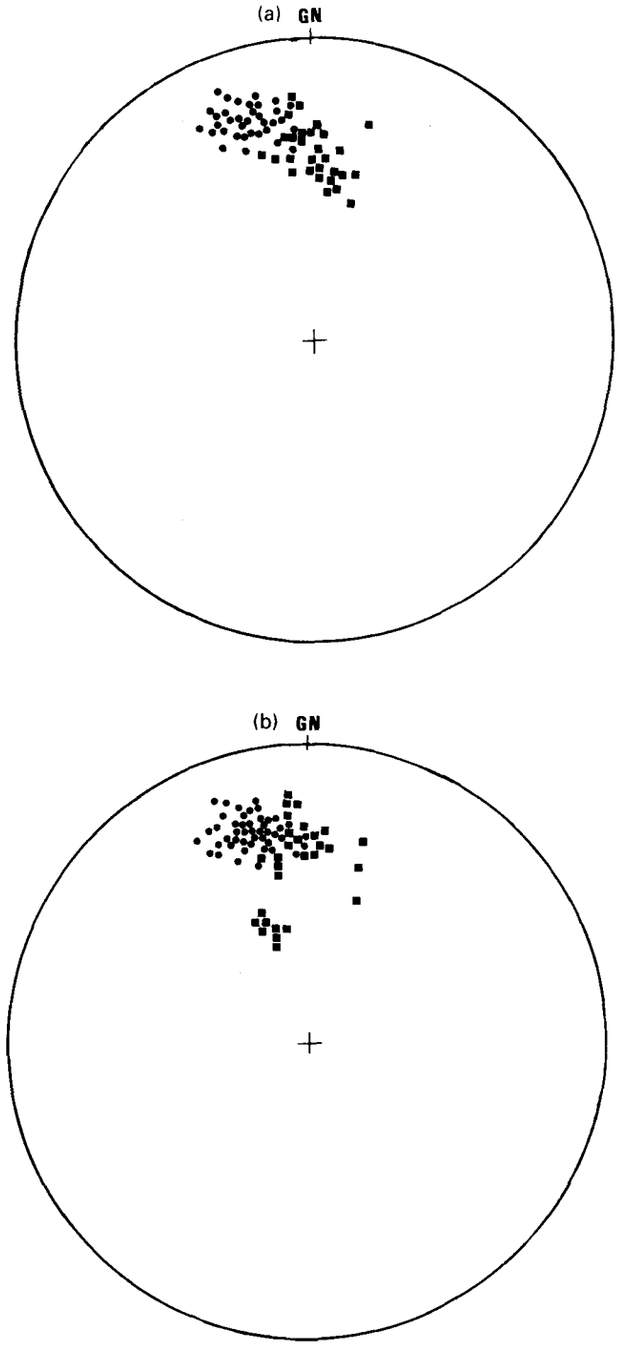


Figure 5. Equal area plot (filled symbols: lower hemisphere) of ChRM vectors before (a) and after (b) dip correction: samples with one (squares) or two (circles) components.

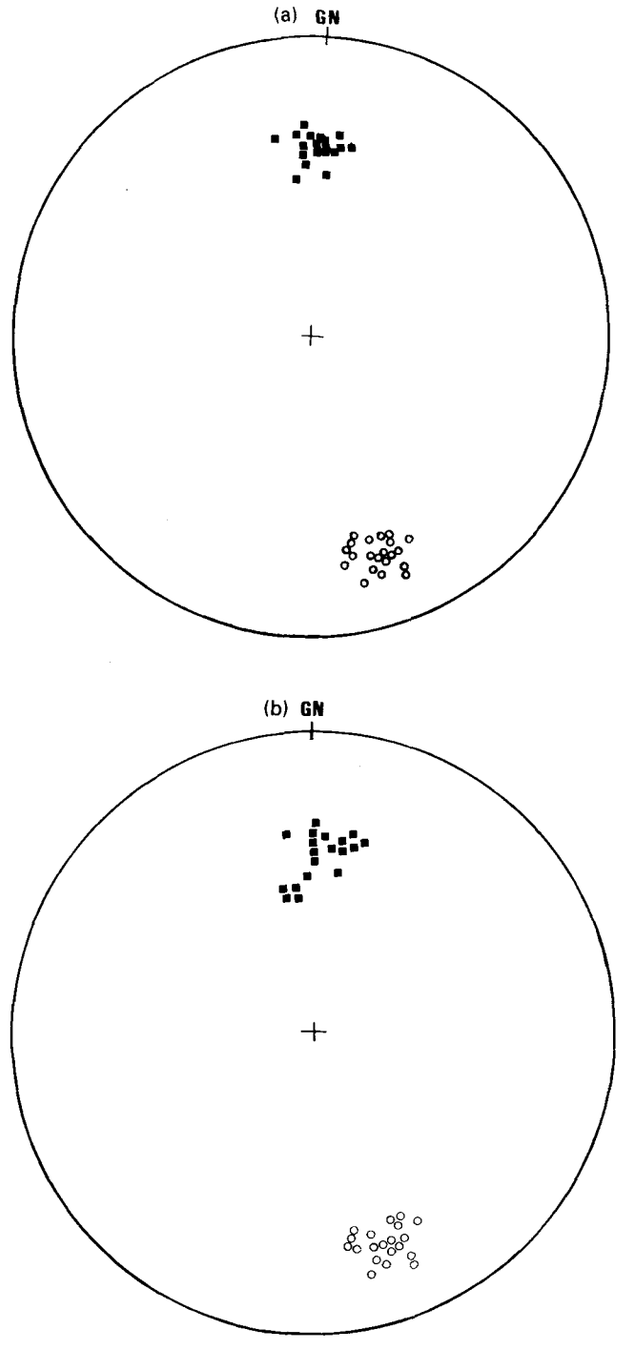


Figure 6. Equal area plot (filled symbols: lower hemisphere; open symbols: upper hemisphere) of Late Triassic–Liasic (circles) and recent overprint (squares) low-temperature directions before (a) and after (b) dip correction.

(samples from site 9 show mainly dominant overprinting). Sites 14 and 2 from, respectively, older and younger levels contain just the normal polarity. However, one sample from the oldest bed of site 14 shows an evolution of magnetization direction from normal polarity towards its opposite at the highest temperature.

The presence of two components in some samples obviously indicates that at least one of these components is the result of a magnetic overprint or of a diagenesis that occurred during two polarity intervals (Merabet & Daly 1986). The existence of two components of normal polarity in the same sample (**group a**) clearly corresponds to different periods of magnetization acquisition, and indicates the overprinting phenomenon; besides, in this case, the fold test is positive for the high-temperature component, while it is clearly negative for the low-temperature one.

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Post-tilting magnetizations

For this **group a**, this last component is therefore related to a partial overprinting limited to some beds (as in the Moscovian beds from the same basin—Henry *et al.* 1992c). The orientation of this second normal component (Table 1) is similar to that of the recent overprint observed in all the Palaeozoic series of this basin (Henry *et al.* 1991). It can be noticed that the 37 rejected ChRM are mostly relatively close to this overprint direction and that the fold test is negative for **group c** as for the low-temperature component of **group a**. We can suggest an explanation of the different magnetic behaviours according to the samples. The recent overprinting should have affected the samples in different ways, those from **group b** not being remagnetized at this time, those belonging to **group a** being partially overprinted (probably obliterating the reversed partial remagnetization) and the 37 rejected ChRM of **group c** being almost wholly remagnetized (superimposition with a weak primary component).

Ante-tilting magnetization

The positive inversion test, for samples with normal ChRM and a reversed second component, shows that the two components were in this case acquired in a relatively short time interval, and the assumption of overprinting or diagenesis during two polarity intervals remains possible in this case. The fact, that the high blocking temperature component is always of normal polarity is consistent with the hypothesis of a primary magnetization, because of the large dominance of the normal polarity during the Late Triassic–Lower Lias period (Hailwood 1989). Similarly, the presence of a weak second anti-parallel component, in some beds, is not surprising if the assumption of diagenesis of long duration is taken into account. Older formations of the same basin underwent only Permian and recent overprints (Henry *et al.* 1991, 1992c) and the single event which specifically affected the Zarzaïtine levels is a rubefaction. The question is then about the age of this rubefaction. Busson & Cornée (1989a,b) have pointed out a syngenetic or diagenetic age for this rubefaction, and the magnetization acquisition age is then almost the same, considering the assumption of a primary magnetization or of a Chemical Remanent Magnetization (CRM) related to rubefaction, i.e. that of deposit and diagenesis.

These results (Table 1) show the existence of at least three periods of reversed polarity during the deposition and the diagenesis of the Zarzaïtine Formation. The first period discovered within the Stegocephalia level (site 9), the second during the Late Triassic (site 14) and the last related to diagenesis during the Lias. The new palaeomagnetic pole for stable Africa at La Reculée ($N = 8$ sites, 70.9°S, 55.2°E, $K = 478$, $A_{95} = 2.3°$), for the Late Triassic–Liasic period, seems therefore well specified as the result of a mean over a long interval of time.

COMPARISON WITH GEODYNAMICAL DATA

The importance of the remagnetization phenomenon in other Triassic formations (Ghorabi & Henry 1991) does not yield a sufficient number of significant palaeomagnetic results for this period, and the two corresponding poles (Table 2) are significantly distinct. The mean Triassic pole position ($N = 2$ poles, 61.5°E, 45.8°S) presently available is different (but within the uncertainty limit) to that determined using attempts at palaeocontinental reconstruction (Table 2) and data from the other parts of Gondwana (Westphal *et al.* 1986). Hence, other palaeomagnetic investigations must be done before any modelling of geodynamical reconstruction, and the present mean pole should be considered only as an indication for comparison with poles for the neighbouring periods.

There are more reliable well-defined and grouped (Fig. 7) results for the Lias, including the La Reculée pole, (poles f–j of Table 2). By taking into account the scattering A_{95} for each of these five sites (Le Goff 1990; Le Goff *et al.* 1992), a Liasic palaeomagnetic pole for stable Africa can be determined: 55.7°E, 69.3°S, $K = 50$, $A_{95} = 2.3°$, $K_y/K_x = 1.06$ (classical Fisher analysis should give the same direction here: $N = 5$ poles, 55.6°E, 69.3°S, $K = 863$, $A_{95} = 2.1°$). The position of this pole is in good agreement with Besse & Courtillot's (1991) synthetic path, but it is outside the confidence limit of Westphal *et al.*'s (1986) poles. The comparison of this pole with corresponding European and North American data (Van der Voo 1990) leads to a classical Wegener palaeocontinental reconstruction A type of Pangea (Fig. 8).

The scattering of the five Dogger poles (Fig. 7) is more important (77.3°E, 64.6°S, $K = 24$, $A_{95} = 4.9°$, $K_y/K_x = 1.32$; classical Fisher analysis: $N = 5$ poles, 77.4°E, 64.4°S, $K = 100$, $A_{95} = 6.2°$), in particular because of pole d (Table 2). The mean pole position without this pole d is 72.4°E, 62.2°S ($N = 4$ poles, $K = 346$, $A_{95} = 3.8°$). Contrary to the Lias pole, this Dogger pole agrees well with Westphal *et al.*'s (1986) synthetic pole, but is outside the confidence limit of Besse & Courtillot's (1991) one. This discrepancy between the different Lias and Dogger synthetic and palaeomagnetic poles shows that these synthetic paths, which are based both on results from the different continents and on assumptions of palaeocontinental reconstruction, need some complementary palaeomagnetic data to be improved.

The Apparent Polar Wander Path (APWP) for stable Africa, the Carboniferous–Autunian part of which was previously improved (Aifa *et al.* 1990; Henry *et al.* 1992a,c), has then a well specified part for the Lias, and, to a less degree, Dogger times. Unfortunately, the Upper Triassic part is insufficiently specified.

If only the positions during the Stephano-Autunian and during the Lias periods are considered, the main *latitudinal* Upper–Palaeozoic–Lower Mesozoic displacement of stable Africa seems to be a northwards drift, with anti-clockwise rotation over 20° (Fig. 9). Northwards drift of the stable Laurasia is similar (Van der Voo 1990), but without significant rotation. This confirms, first, a global displacement of Pangea during this period of transition from the A2 to A type (Van der Voo & French 1974). African Triassic poles seem to indicate that the rotation probably ended before the Upper Triassic. Considering North America as a fixed reference frame, displacement corresponded to a rotation of 24° (pole of rotation at 13°N and 42°W for North America located in its present position). Such an important rotation would have had some effect, at least in the Hercyno-Appalachian collision area and probably in

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Table 1. For each cluster of ChRM in each site (* indicates presence of both polarities in the site), number of directions (N), mean direction (D, I) before and after dip correction. Mean values for each cluster and palaeomagnetic pole (lat., long.) before and after dip correction, respectively, for recent overprint and Late Triassic—Liasic ChRM. Corresponding Fisher's (1953) parameters (k, α_{95}, K, A_{95}).

		before dip correction				after dip correction							
site	N	D(°)	I(°)	k	α_{95}(°)	D(°)	I(°)	k	α_{95}(°)	lat. (°S)	long. (°E)	K	A_{95}(°)
recent overprint													
2	8	7.0	38.6	164	3.9	7.0	38.6	164	3.9				
9	(11	4	37			348	51		D and I from remagnetization circles analysis)				
13	6	1.6	37.5	257	3.6	2.4	39.2	257	3.6				
14	5	356.8	38.2	149	5.1	356.8	38.2	149	5.1				
mean	19	2.6	38.2	150	2.6	(2.8	38.8	151	2.6)				
4 sites		2.3	37.9	547	3.0	(359.2	42.0	92	7.3)	83.0	351.3		
4 sites		-	-	-	-	-	-	-	-	83.3	348.7	408	3.5
Late Triassic - Liasic ChRM													
2	3	346.3	30.1	188	5.9	346.3	30.1	188	5.9				
4	2	346.4	23.2	-	-	348.5	22.5	-	-				
5*	10	345.1	26.3	200	3.1	347.5	25.8	200	3.1				
6*	14	341.0	21.0	294	2.2	342.9	20.9	294	2.2				
8*	9	349.8	27.7	134	4.0	352.3	26.7	134	4.0				
10	8	343.7	23.7	111	4.7	343.8	27.7	111	4.7				
13*	11	342.7	26.2	114	3.9	343.0	28.1	114	3.9				
14	11	343.5	28.3	221	2.8	343.5	28.3	221	2.8				
mean	68	344.1	25.5	135	1.5	345.3	26.0	126	1.5				
8 sites		344.8	25.8	442	2.4	346.0	26.3	353	2.6	70.9	55.1		
8 sites		-	-	-	-	-	-	-	-	70.9	55.1	478	2.3

north-eastern Africa, where this rotation implies a more important northwards drift. In fact, Permo-Triassic dextral movements, along structures more or less parallel to the limit of Africa and Laurasia, have been observed in the Hercyno-Appalachian collision zone, for instance, in Iberia (Sopena *et al.* 1988), Morocco (Mattauer, Proust & Tupponier 1972; Ricou 1991) and the North American margin (Reed & Bryant 1964; Rast & Grant 1977). Just east of this collision area, the opening of some basins (Lago Negro–Sosio; Ricou 1991) occurred during the Early Triassic, even perhaps during the Permian, thus implying distension. More easterly, north of the Arabian promontory, some microcontinents (like Central Iran), which were parts of Gondwana at least until the Upper Palaeozoic, were aggregated to Laurasia during the Upper Triassic (Belov, Gatinsky & Mossakovsky 1986); traces of subduction, under Laurasia, on their northern margins are shown by Permo-Triassic magmatism of the Hindu Kuch and Ciscaucasia (Dercourt *et al.* 1986; Ricou 1991); this northwards drift of microcontinents could be in fact not only related to the opening of the Neotethys, but also connected, at least in the beginning, to the rotation of Africa.

During Dogger times, when Laurasia, which was relatively stable (Van der Voo 1990), probably underwent a weak clockwise rotation (Savostin *et al.* 1986), the African APWP contains indications of a similar clockwise rotation of Africa (of the order of 10°) which is in agreement with previous geological models (Savostin *et al.* 1986). Such a rotation is probably related to the major event for Laurasia and Gondwana during the Jurassic, the beginning of the Atlantic opening.

CONCLUSIONS

The NRM of the Zarzaïtine Formation of the La Reculée area is composed, according to the samples, of superimposed or juxtaposed magnetizations. In many cases, the separation of these magnetization components was possible, and a Late Triassic–Liasic ChRM of normal polarity has been obtained. In addition, a second component corres-

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Table 2. List of selected and 'synthetic' African poles for Triassic to Dogger. (*) Refers to the GJRAS pole list (McElhinny 1968, 1969, 1972; McElhinny & Cowley 1977, 1980). (+) Absolute age according to Odin & Odin (1990) time-scale.

	Age	Rock Unit	lat (°S)	long (°E)	Ref.
a	140-178	Nigeria	62	62	16/120(*)
b	161-173	Hoachanas	62	72	14/250(*)
c	154-190	Karroo	65	75	8/59(*)
d	173-181	Marangudzi	71	107	10/77(*)
e	173-181	Mateke	59	80	8/63(*)
f	173-192	Liberia	69	62	14/248(*)
g	180-186	Draa	66	50	13/36(*)
h	182-192	Hank	69	52	Sichler et al., 1980
i	182-192	Hodh	71	60	Sichler et al., 1980
j	180-210(+)	La Reculée	71	55	this study
k	205-230(+)	Zambia red sandstones	68	50	8/67(*)
l	210-230(+)	South Tunisia	55	43	Ghorabi and Henry, 1991
Mean poles					
154-180(+) - Dogger					
		(a-e)	64	77	
		(a, b, c, e)	62	72	
180-205(+) - Liasic					
		(f-j)	69	56	
205-230(+) - Upper Triassic					
		(k-l)	61	46	
"Synthetic" poles					
154-180(+) - Dogger			60	72	Westphal et al., 1986
180-205(+) - Liasic			67	72	Westphal et al., 1986
205-240(+) - Middle-Upper Triassic			64	58	Westphal et al., 1986
240-245(+) - Lower Triassic			53	54	Westphal et al., 1986
	160		55	65	Besse and Courtillot, 1991
	170		63	52	Besse and Courtillot, 1991
	180		65	54	Besse and Courtillot, 1991
	190		68	63	Besse and Courtillot, 1991
	200		66	63	Besse and Courtillot, 1991

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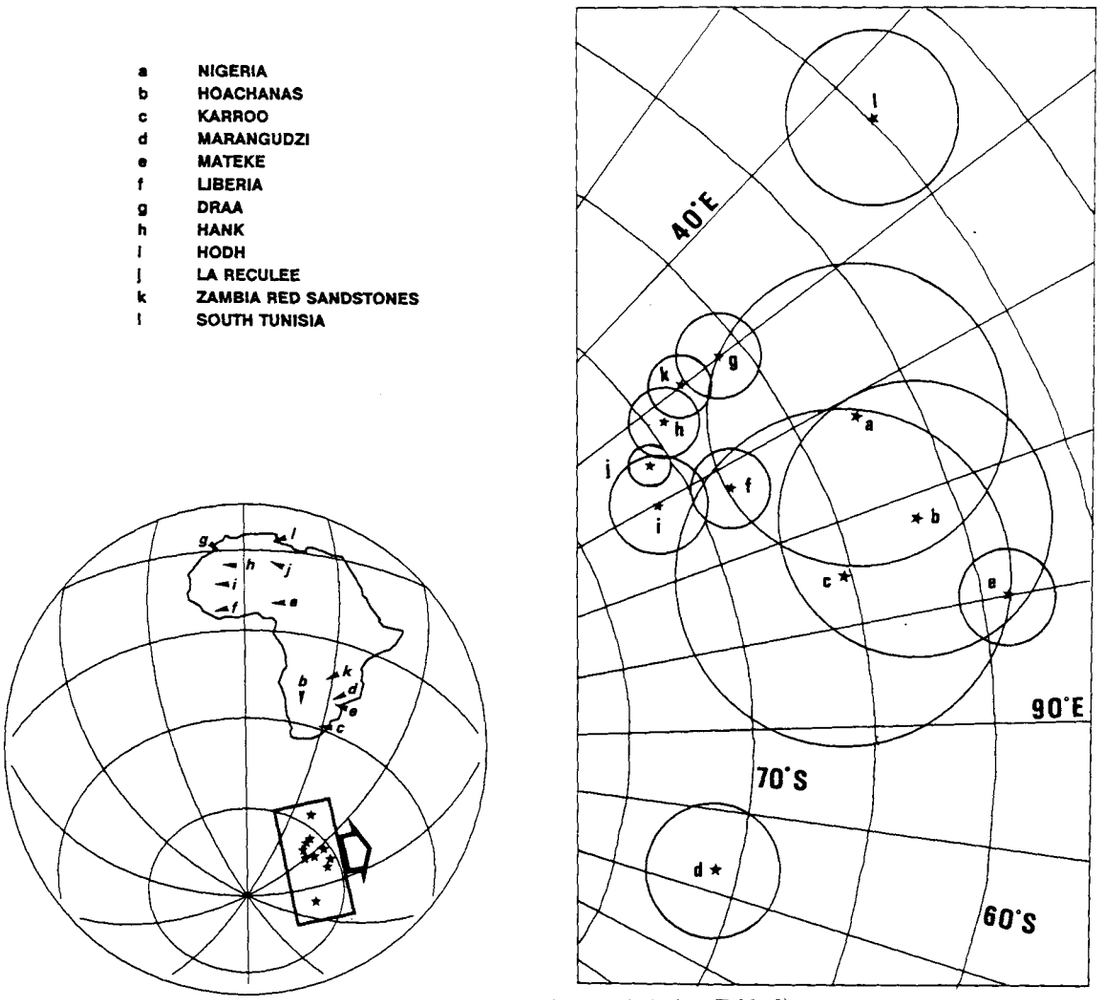


Figure 7. African Triassic–Dogger palaeomagnetic poles with A_{95} confidence circle (see Table 2).

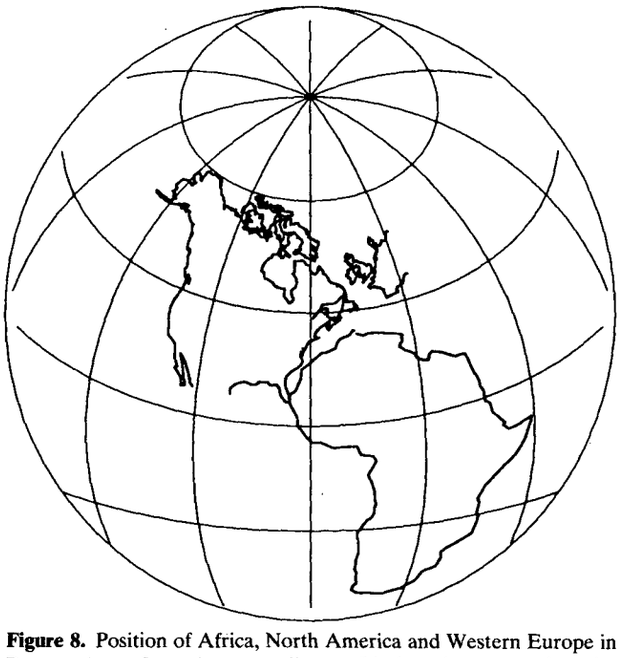


Figure 8. Position of Africa, North America and Western Europe in Pangea A configuration according to Liasic palaeomagnetic poles (Van der Voo 1990; this study).

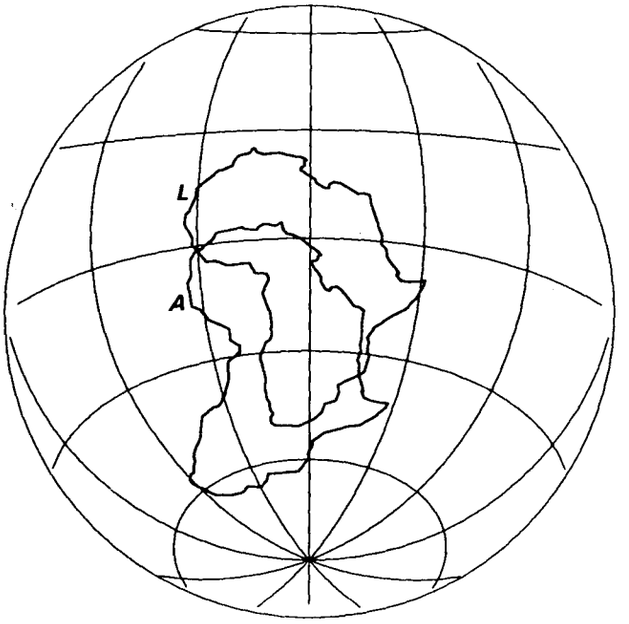


Figure 9. Latitudinal displacement of Africa from Autunian (A) to Late Triassic–Lias (L).

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ponding either to another Late Triassic–Liasic reversed polarity magnetization or to a recent overprint has also been identified. The new Late Triassic–Liasic palaeomagnetic pole confirms a latitudinal displacement, with an important anti-clockwise rotation, of stable Africa towards the north from the Autunian to Lias. A corresponding northwards drift of Laurussia occurred without significant rotation, and this evolution, which corresponds to the transition of the Pangea configuration from A2 to A, ended before the Late Triassic.

ACKNOWLEDGMENTS

We are very grateful to the Algerian DPRS, the French Foreign Office and the SONATRACH (especially Dr Attar and Dr Hamel, and the geological staff of In Amenas) which greatly supported this work, to G. Busson and L. E. Ricou for helpful discussions, and to M. Le Goff, H. Harchaoui and R. Bendjelloul for assistance in the field.

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